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DESCRIPTIVE ECONOMICS RESEARCH PAPER

Financing Public Transport and Free-Fare Experiments

RECONCILING SERVICE ACCESSIBILITY AND GREENHOUSE GAS
REDUCTION THROUGH FREE PUBLIC TRANSIT

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Contents

Introduction	2
1 Overview of the Public Transport Sector: an Industry Shaped by Crises and Political Trends	4
1.1 A diverse set of actors for an essential sector	4
1.2 Major structural trends with distinct objectives for future challenges	5
1.3 A carbon-intensive transport sector where public transit struggles to compete with private vehicles	6
2 Assessing the Relevance of Free Fares: Ecological, Social, and Economic Perspectives	9
2.1 Environmental, social, and economic benefits of free fares	9
2.2 The financing model for public transport	11
2.3 Impacts on the traditional financing model	12
3 The Effective Impact of Free Fares: Limits, Modalities, and Implementation	14
3.1 Limits of free fares	14
3.2 Different modalities of free-fare implementation	15
Conclusion	17
Bibliography	18

Introduction

A study of the history of collective transport in Europe reveals a complex tapestry in which questions of accessibility and environmental preservation have been closely intertwined. At the turn of the nineteenth century, with the onset of the industrial revolution in the West, public transport systems emerged as a response to the rapid growth of cities. Today, when we speak of urban public transport (UPT), we refer to buses, trams, metros, and in the Paris region to commuter trains and the RER network. Trams and buses, initially powered by horses, then by steam engines and finally by electricity, allowed a growing share of citizens to access expanding urban centres. This considerably improved access to services, but the expansion also brought its share of problems, including congestion, air pollution, and noise.

Over the course of the twentieth century, the private car progressively came to dominate the urban landscape, steadily eroding the modal share of public transport. By 2020, car journeys accounted for 70% of trips in major cities. Private vehicles had become synonymous with freedom of movement, but they also exacerbated environmental problems. Growing concerns over air quality and climate change have, however, rekindled interest in sustainable public transport. Many transport networks have progressively adopted cleaner and more efficient technologies, while seeking to improve accessibility for all citizens.

The idea of free public transport also has a complex history in the Western world since the mid-twentieth century. It was introduced in certain European and American cities from the 1950s onwards, primarily for economic or political reasons. The United States was among the first to experiment with free collective transport: the first city in the world to make transit free was Commerce (population 12,947 in 2017), in the suburbs of Los Angeles, in 1962. Over time, however, free fares were abandoned in many places in favour of targeted social pricing. The Belgian city of Hasselt, for instance, had adopted free public transport in 1996 with undeniable success, only to reverse the policy in 2014 in order to fund network improvements. More recently, renewed interest in free fares has emerged in the West, not only to improve accessibility but also as a potential tool to reduce car dependency and cut greenhouse gas (GHG) emissions.

Free public transport raises crucial questions about how it might address contemporary challenges — service accessibility, energy concerns, and GHG reduction — while responding to citizens' evolving mobility needs and preserving the environment. This calls for an examination of how the lessons of the past can illuminate future solutions, offering a glimpse of how free public transit could be a tool for a more sustainable and inclusive future.

Free public transport is frequently put forward as a solution to these challenges. However,

this measure is far from universally accepted and lies at the heart of many debates. While it is presented by some as an ecological measure giving everyone access to mobility, others regard it as unrealistic. The central question is therefore:

To what extent can making public transport free reconcile service accessibility with a reduction in greenhouse gas emissions?

To answer this question, we will first take stock of the public transport sector: its actors, its structural trends, and its attractiveness relative to the private car. We will then explore the relevance of free fares from ecological, social, and financial perspectives, examining existing financing models and possible alternatives. Finally, we will analyse the actual impact of free-fare policies through documented experiences, confronting theoretical promises with observed empirical results, with all the methodological caution that entails.

Overview of the Public Transport Sector: an Industry Shaped by Crises and Political Trends

A diverse set of actors for an essential sector

The public transport system in France rests on a collaboration between several key actors, each playing a specific role in the organisation, management, and regulation of the network.

On the legislative side, the French State is the primary actor: it plays a central role in defining transport policy, allocating infrastructure funding, and overseeing the sector as a whole. It sets the broad directions and objectives for sustainable mobility. The Mobility Orientation Law (LOM) adopted in 2019, for example, aims to transform France's mobility system by emphasising several key objectives: reducing greenhouse gas emissions, improving accessibility and equity, accelerating the energy transition, and promoting public transport.

Transport organisation is handled at the level of local authorities — cities, départements, and regions — through Mobility Organising Authorities (MOAs, known in French as AOM). Under the law, MOAs “organise regular public transport services and may organise on-demand transport services. They contribute to the development of non-motorised land transport modes and shared uses of motor vehicles.” MOAs select operators to run the network, collect revenue, and invest. They draw up contracts with transport operators, set fares and routes, and oversee service quality. A very small number of operators share the entire French market — notably Keolis and Transdev — making the urban transit sector an oligopoly.

The Paris region (Île-de-France) is a particular case, concentrating the bulk of urban transit demand in France. The network's organising authority, Île-de-France Mobilités (formerly STIF), delegates management to several bodies:

- The Régie Autonome des Transports Parisiens (RATP), a public company operating metro, bus, parts of the RER A and B lines, and tram lines within Paris and its inner suburbs, accounting for 72% of journeys;
- The national rail operator SNCF, which runs suburban rail (Transilien) and certain RER segments, accounting for 20% of journeys;
- Optile, a professional association grouping nearly a hundred private bus companies operating mainly in the outer suburbs, accounting for 8% of journeys.

Finally, passengers themselves play an active role in the system. Their needs, preferences, and feedback are regularly taken into account to improve service quality and user satisfaction.

Major structural trends with distinct objectives for future challenges

The evolution of transport is intrinsically tied to major trends, shaped in recent years by the disruptions caused by the Covid-19 pandemic. The rise of remote working significantly reduced travel demand, prompting a reflection on transport accessibility for peri-urban populations.

For decades, population growth has gone hand in hand with urban sprawl, generating an ever-increasing need to extend the geographic coverage of public transport networks. This urban sprawl is clearly visible in the data: between 2000 and 2010, the share of urban territory grew by just under 20%. The example of the Lyon metropolitan area is illustrative: the urban footprint expanded dramatically at the expense of rural areas between 1955 and 2011.

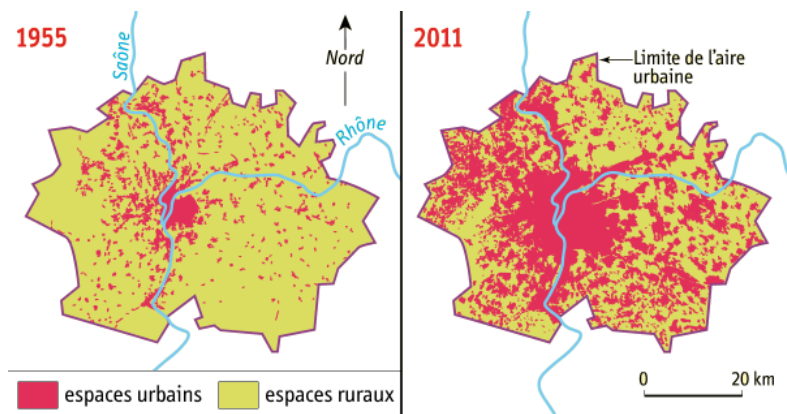


Figure 1: Expansion of the Lyon urban area between 1955 and 2011.



Figure 2: Share of the urban population in France (%), 1936–2020.

Source: Centre d'observation de la société, 2023

According to INSEE, nearly 80% of the French population lived in an urban area in 2020.

Public policy has aligned with the goal of expanding public transport coverage. Between 1995 and 2015, the population living in administrative areas of more than 50,000 inhabitants grew by 30.6%. In 2018, the national census recorded more than 330 territorial service areas (RT) across France. Despite this expansion, however, Cerema estimated in 2018 that 27% of the French population still lived outside an area served by a MOA.

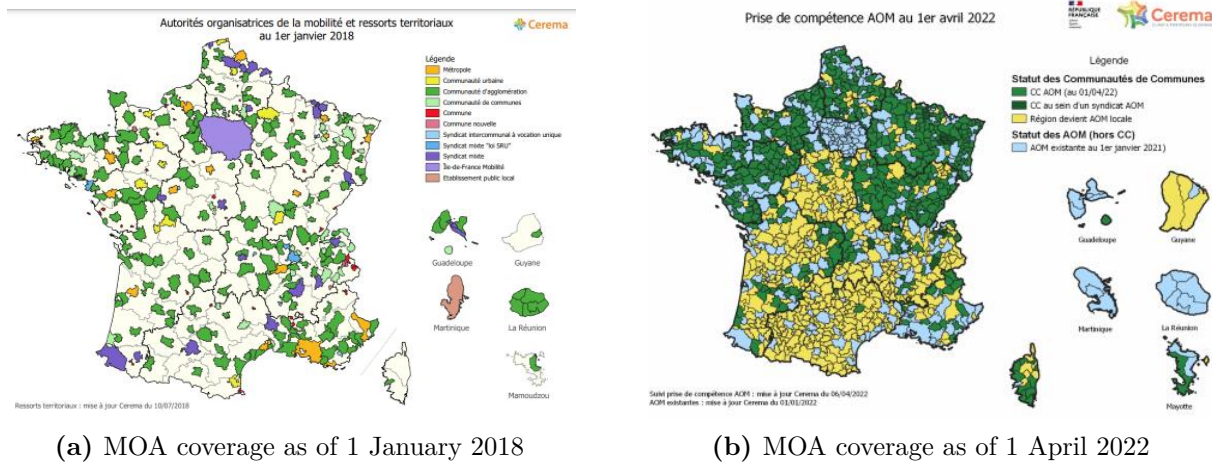


Figure 3: Evolution of MOA territorial coverage in France.

Sources: Cerema, 2018; Cerema, 2021

A carbon-intensive transport sector where public transit struggles to compete with private vehicles

The public transport sector has been heavily affected by major societal crises. The most striking recent example is the Covid-19 pandemic. During lockdown periods, transit ridership fell to between 5% and 10% of normal levels, with a more than 30% decline sustained throughout 2020.

In 2018, transport accounted for 14.3% of household expenditure, with 82.8% of that figure attributable to private vehicle use. The table below summarises household transport consumption in 2018.

Table 1: Household transport consumption in 2018 (value).

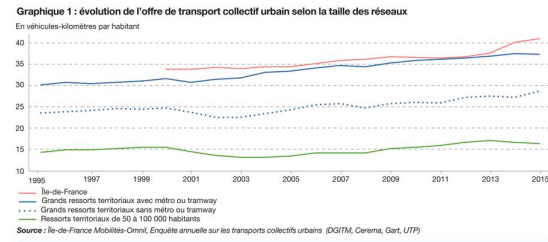
Source: INSEE, National Accounts, base 2014, May 2019. Calculations: SDES

	Level 2018 (bn)	Change 2016 (%)	Change 2017 (%)	Change 2018 (%)
Private transport	143,851	3.4	6.4	6.3
Vehicle purchase	43,809	7.7	5.4	2.0
Running costs	100,042	1.4	6.9	8.3
Public transport	29,980	0.0	6.3	2.5
Total transport expenditure	181,761	2.8	6.4	5.6
Total household consumption	1,219,756	2.0	2.2	2.4

According to the International Energy Agency (2016), the transport sector accounts for nearly a quarter of global CO₂ emissions, with road traffic responsible for close to three quarters of that share. In France, while overall GHG emissions fell by nearly 20% between 1990 and 2018 to reach 445 million tonnes of CO₂ equivalent, GHG emissions from the transport sector actually *increased* by 9% over the same period. The passenger transport sub-sector is the single largest emitter, responsible for more than 30% of total GHG emissions and over 40% of CO₂ emissions.

Niveau en milliards d'euros	Niveau	Évolution annuelles		
Évolution en %	2018	2016	2017	2018
Déplacements individuels	143 851	3,4	6,4	6,3
Achat de véhicule	43 809	7,7	5,4	2,0
Dépenses d'utilisation	100 042	1,4	6,9	8,3
Déplacement en TC	29 980	0,0	6,3	2,5
Dépenses de transport	181 761	2,8	6,4	5,6
Consommation des ménages	1 219 756	2,0	2,2	2,4

(a) GHG emission breakdown in 2018.
Source: CITEPA, Secten report 2019



(b) GHG emission trends in the transport sector, France, 1990–2019.
Source: CITEPA, Secten report 2020, SDES, 2021

Figure 4: GHG emissions from the French transport sector.

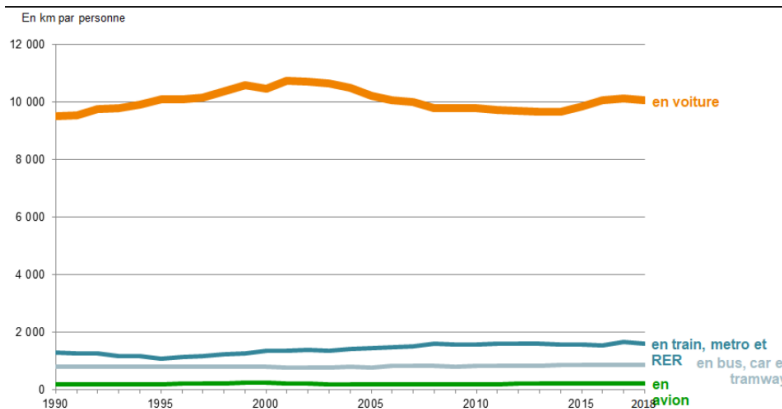
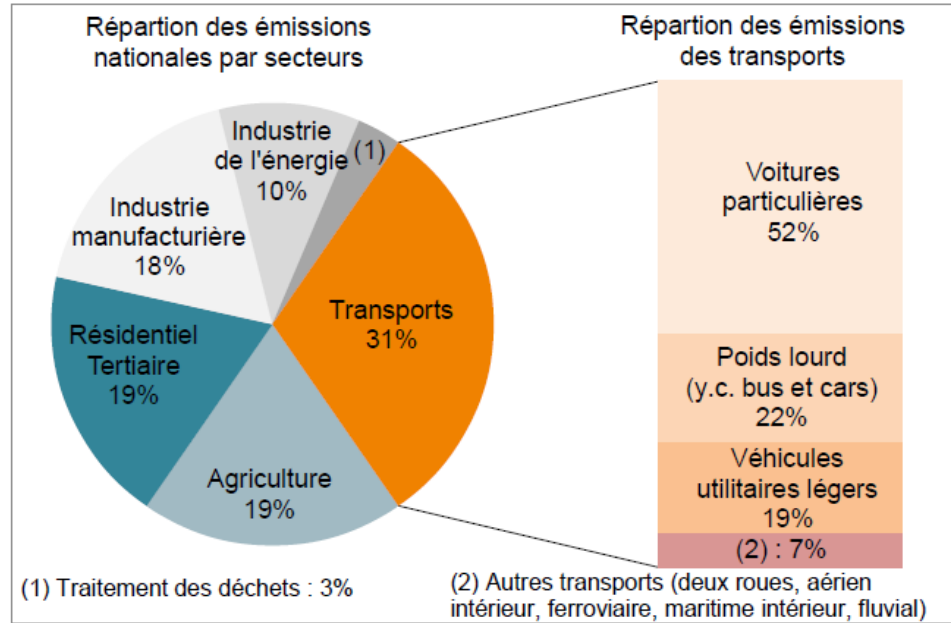


Figure 5: Trends in average annual distance travelled per person, France, 1990–2018.
Source: INSEE 2020, SDES

A useful indicator of supply trends is the kilometric offer, which measures the total vehicle-kilometres produced annually per inhabitant covered. In areas of more than 50,000 inhabitants outside the Paris region, this indicator grew steadily from 25.7 vehicle-km/inhabitant in 1995 to 30.2 in 2015.

En %



Champ : France métropolitaine et DROM (périmètre Kyoto).

Source : Citepa, rapport Secten 2019

Figure 6: Evolution of urban public transport supply by network size (vehicle-kilometres per inhabitant).

Source: Île-de-France Mobilités-Omnîl, Annual survey on urban public transport (DGITM, Cerema, Gart, UTP)

Assessing the Relevance of Free Fares: Ecological, Social, and Economic Perspectives

Environmental, social, and economic benefits of free fares

While free fares are primarily motivated at first glance by social concerns — ensuring that even those on modest incomes can access public transport — operational experience shows that they also serve objectives of urban restructuring and network optimisation, particularly to address the problem of underutilised vehicles. Environmental motivations are also at play, with the aim of reducing private car use.

From a historical perspective, free fares have emerged in a context of broader change in urban transit. Four broad phases can be identified:

- The pioneer phase (1971–1999);
- The expansion phase from 2000 to 2011, with a growing number of small cities introducing free fares;
- The consolidation phase from 2011 to 2016, during which few cities joined or left the free-fare movement. The most instructive international example of this period is Tallinn, the Estonian capital with over 400,000 inhabitants, which introduced free fares in 2013. The Tallinn case is instructive on two counts: first, it demonstrated that free fares need not be confined to small networks, thereby opening up the debate in larger European cities; second, post-implementation evaluations revealed that the ridership increase (roughly +10%) primarily benefited lower-income residents and elderly passengers rather than car drivers, which raises questions about the capacity of free fares alone to alter modal split;
- The renewal phase from 2017 onwards, with medium-sized cities such as Dunkirk. In 2023, the Montpellier metropolitan area (450,000 inhabitants) became the largest local authority to offer free public transport to its residents.

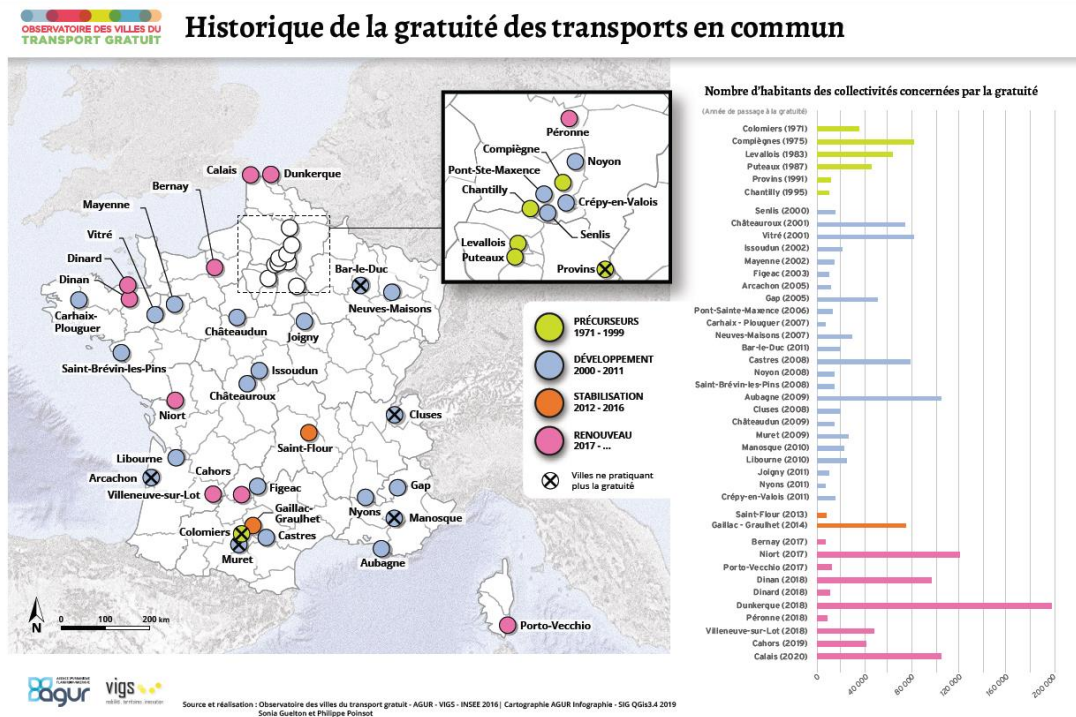


Figure 7: Historical timeline of free-fare public transport in France.

Source: “Free urban transit networks: a particular financing model?” Sonia Guelton and Philippe Poinot

On the question of equity and redistribution, the context of widening income inequality is important. Free fares directly increase the purchasing power of passengers, with the gain being proportionally much more significant for lower-income users. For higher-income users, the idea is that this purchasing power boost may be redirected towards local commerce, linking social and environmental benefits simultaneously. Beyond the equity dimension, a practical benefit also emerges: under means-tested pricing systems, a significant non-take-up phenomenon exists, whereby eligible individuals fail to claim the subsidised fares they are entitled to — due to administrative complexity, lack of information, or fear of stigmatisation. Cécile Féré, an urban planning researcher at the Société Publique Locale de Lyon, found that only a third of the 36,000 potential beneficiaries of reduced fares in Lyon were actually using them in the early 2000s, compared to 40% in Nancy and 50% in Nantes. In the Paris region, the non-take-up rate was estimated at over 70%.

Another important argument concerns the underutilisation of public transport networks. According to Niort’s mayor Jérôme Baloge, an underused network amounts to “public money converted into CO₂.” Indeed, before the introduction of free fares in Niort, 25% of bus journeys were made with fewer than five passengers on board.

The resulting increase in transit ridership can in part be explained by reduced car use. In small cities, the effectiveness of free fares is documented. In Aubagne, between 2009

and 2013, public transport ridership increased by 170% while car traffic fell by 10%. In Dunkirk, the first large French city to adopt free fares, bus ridership rose by 70% on weekdays and by 140% at weekends.

It is important, however, to interpret these figures with caution. These are correlational observations: the decision to introduce free fares is rarely made in isolation from other simultaneous reforms (network redesign, new vehicles, communication campaigns), making it difficult to attribute causally the ridership increase to pricing alone. In Dunkirk, for instance, free fares were introduced alongside a High-Level Service Bus network. Furthermore, cities that opt for free fares are not a representative sample: they typically already have low fare revenues and a strong tradition of social pricing, which introduces a selection bias in the interpretation of results. Without a rigorous counterfactual, the positive figures observed cannot be taken to imply that free fares would be equally effective in different contexts, particularly in large metropolitan areas.

The financing model for public transport

Urban public transport has a traditional financial architecture: it rests on a contract between MOAs (which build the network and define objectives) and public or private operators (which maintain and run it). The financing of UPT comes from three main sources: fares paid by passengers, a payroll tax levied on certain employers known as the *versement mobilité* (mobility levy, or ML), and contributions from local authorities and the State.

Outside the Paris region, the ML provides just under 50% of UPT financing, while the user share remains below 20%. In Île-de-France, the user share is higher, reaching 38%.

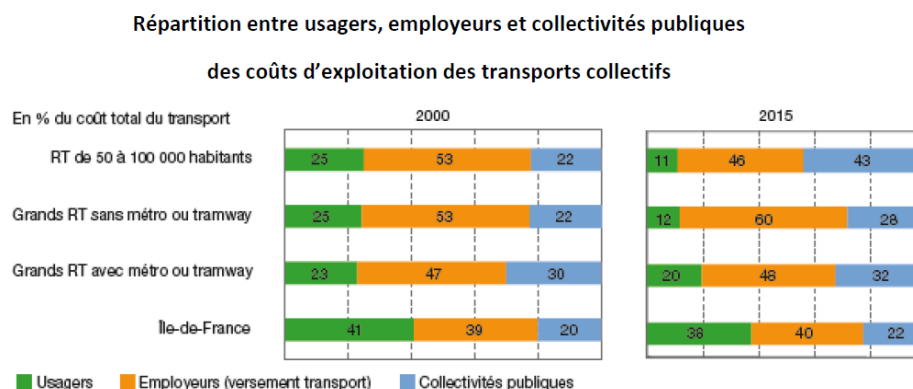


Figure 8: Breakdown of urban public transport operating costs between users, employers, and public authorities (2000 and 2015).

Source: Guillaume Gontard, Senate Information Report, 2019.

Table 2: Evolution of MOA revenues outside the Paris region (in millions of euros).
Source: DGITM responses to parliamentary questionnaire

	2015	2016	2017	2018	2019	2020	Share 2020
Mobility levy	3,753	3,827	3,875	4,080	4,287	4,037	42%
Local authorities	2,725	2,760	2,833	2,817	3,082	3,079	32%
Fare revenues	1,316	1,325	1,379	1,448	1,494	1,010	11%
State	104	85	105	122	164	680	7%
Borrowing	703	594	615	1,138	976	706	7%
Total	8,601	8,591	8,807	9,605	10,003	9,512	100%

While the mobility levy has been the privileged financing instrument for UPT in France for forty years, it is no longer sufficient to cover investment and operational needs. Having reached its ceiling in most MOAs and in a context of weakening payroll bases, the resulting revenue shortfall raises a critical question: how to compensate?

Between 2000 and 2015, fare revenues across provincial networks grew from 0.8 billion to 1.3 billion. However, the coverage of operating costs by fare revenues has been declining: it fell from approximately 30% in 2000 to 18% in 2015 in networks serving between 50,000 and 100,000 inhabitants.



Figure 9: Trends in operating costs, fare revenues, and ridership.

Source: “Urban public transport: despite rising operating costs, passenger financial contribution is declining”

Impacts on the traditional financing model

The question of free fares is therefore not merely about who benefits from the system, but fundamentally about who finances it.

Two main characteristics tend to distinguish cities that adopt free fares. First, free fares are typically found in small networks with a limited number of lines: mostly cities with fewer than 50,000 inhabitants. In 2018, 20 of the 29 local authorities offering free fares operated a network of five bus lines or fewer. Second, free fares tend to target networks

with already low fare revenues. In such cases, the costs associated with ticketing — vending machines, validators, sales staff, inspection agents — may represent a significant share of operating costs, making their elimination financially attractive. The national average user contribution to operating costs is around 25%, whereas in networks that have introduced free fares it was below 15% (12% in Niort) or even around 10% (Dunkirk).

New financing mechanisms may be required to compensate for lost fare revenues. Four main alternatives have been identified:

1. **Urban congestion charges.** Several European and global cities have already implemented such schemes, simultaneously discouraging car use and generating revenue for public transport. However, in the still-recent political context of France’s “gilets jaunes” (Yellow Vest) movement, any additional financial burden on car users carries significant political risk.
2. **Land value capture.** Transport infrastructure generates real estate value uplift, and two main options exist to recover part of these gains: taxing the potential capital gain attributable to the infrastructure project, or taxing the capital gains actually realised at the point of sale. Both options present practical difficulties and may create perverse effects on housing market liquidity.
3. **Parking charges,** particularly at shopping centres, offer a dual benefit: discouraging car use and generating revenue for public transport.
4. **Earmarked national tax revenues,** exploiting the principle of budgetary flexibility to redirect a portion of fiscal receipts towards free public transit.

The Effective Impact of Free Fares: Limits, Modalities, and Implementation

Limits of free fares

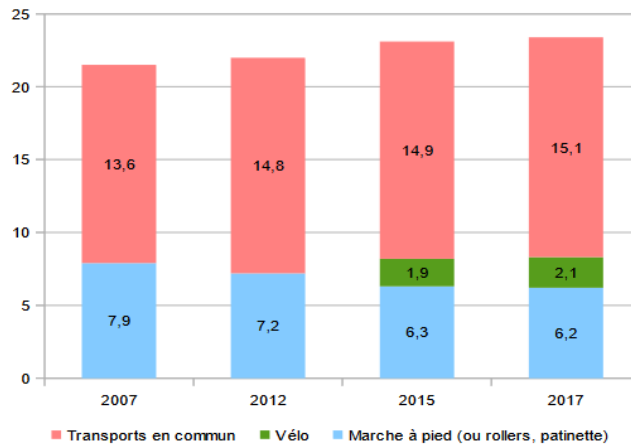
While free fares clearly carry undeniable social and economic merits, their limitations are pointed out by critics. Jean Sivardière, vice-president of the national federation of public transport user associations (Fnaut), which is publicly opposed to free public transport, noted: “In small urban areas, the free public transport offer remains low-grade.” This critique points to a genuine risk: in the absence of sufficient fare revenue to finance service improvements, free fares can paradoxically degrade the offer and produce the opposite of the intended effect.

The central question remains that of modal shift — that is, the capacity of free fares to convince car drivers to abandon their vehicles. The available data are convergent here, but not favourable to a strong ecological case. Frédéric Héran, an urban economist at the University of Lille, estimated that modal transfer from car to public transport amounts to only 1–2%, compared to 2–4% from pedestrians and 5–7% from cyclists. This estimate is consistent with the findings of the 2019 household survey conducted in Dunkirk, in which 24% of new users reported having replaced a car journey with a bus journey — an encouraging figure, but one that covers only a fraction of the additional trips generated. According to modelling by Île-de-France Mobilités, between 60% and 75% of new public transport trips would in fact replace journeys previously made on foot. Free fares thus increase ridership, but their net impact on GHG emissions is limited: they reduce walking and cycling, not car travel.

This tension between the favourable figures observed in small cities and the more pessimistic estimates for large agglomerations is partly explained by structural differences. In a small city with a dense bus network, the car is often the direct alternative to public transport. In a large metropolis, the private car is already poorly competitive for many journeys due to congestion and parking costs; the marginal gains from zero pricing are therefore smaller. This considerably qualifies the ecological scope of the measure: free fares are not a universal decarbonisation lever, but a tool whose effectiveness depends closely on local context.

Figure D1-8 Indicateur de suivi SNBC - Part des déplacements en transport en commun et mode doux (marche, vélo) dans les déplacements domicile travail

En %



Source : Insee, enquêtes annuelles de recensement

Figure 10: Share of public transport and soft mobility modes in commuting trips.

Source: INSEE, Annual census surveys

Furthermore, the desired increase in UPT use requires an increase in operating expenditure to maintain service quality and cope with potential incivilities. In Grenoble, Lille, and above all in Paris, estimates suggest that free fares would cost respectively more than 50 million, 100 million, and over 3 billion per year. These costs sometimes equal the total investment budgets of transport networks, raising serious concerns about the financial viability of the measure. Dunkirk itself chose to forgo a planned tram project in order to free up financing for free fares: this represents a real trade-off between two legitimate objectives (pricing accessibility and infrastructure improvement) whose long-term consequences for MOA investment capacity deserve careful consideration.

It is important to bear in mind that free fares are, above all, a political choice. By introducing free fares, collective transport services become universal, eliminating all user segmentation. As Jean-Louis Sagot-Duvaurox notes, however, this logic leads to the question: “Why transport rather than school canteens?” This question does not invalidate the case for free fares, but it is a reminder that public resources are finite, and that a euro invested in free fares is a euro that cannot fund network expansion, journey-time reductions, or the decarbonisation of the vehicle fleet — potentially more powerful levers for GHG emissions.

Different modalities of free-fare implementation

Free fares can take four main forms:

1. **Socially targeted free fares** (means-tested or age-based pricing);
2. **Geographically targeted free fares**, for example making a single line free;
3. **Temporally targeted free fares**: during summer periods or peak pollution episodes;
4. **Universal free fares** covering the entire network and all users.

Most networks already offer discounted fares for minors and students. Some MOAs also provide reduced or free fares for job-seekers, as in the case of Bordeaux Métropole. Targeted social pricing, as implemented by cities such as Strasbourg and Clermont-Ferrand, effectively addresses the need for inclusive mobility without compromising the financial balance and investment capacity of MOAs.

Free fares during pollution peaks are widespread. The French Environmental Code states that in the event of restrictions on certain categories of vehicles, “access to public passenger transport networks shall be ensured by any pricing measure decided by the transport organising authorities, or free of charge.”

The Dunkirk case. The introduction of universal free fares in Dunkirk in 2018 was accompanied by a comprehensive overhaul of the transport offer, including the introduction of High-Level Service Buses. The revitalisation of public transport thus operated simultaneously on the supply side (new network design) and the demand side (free fares), as part of a broader urban mobility project called “DK Plus de mobilité,” which also included the redesign of several public spaces in Dunkirk city centre. At the time of the transition to free weekend fares in 2015, fare revenues accounted for only 10% of the network’s operating costs, and half of all trips were already free due to social pricing. Between June 2018 and June 2019, ridership increased by 85% on weekdays and by 120% at weekends.

Conclusion

The analysis conducted throughout this paper enables a nuanced answer to the question posed in the introduction: *to what extent can making public transport free reconcile service accessibility with a reduction in greenhouse gas emissions?*

On the accessibility dimension, the answer is clearly positive for small and medium-sized networks. Free fares eliminate the non-take-up phenomenon for social pricing, reduce mobility inequalities, and significantly improve ridership — as documented in Aubagne, Dunkirk, and Tallinn. These gains are real, well-documented, and consistent with the social equity objectives that typically motivate this policy. Free fares do not, however, resolve geographical accessibility inequalities: an area that is poorly served by public transport will remain poorly served, even if fares become free.

On the environmental dimension, the conclusions are more ambiguous. The available evidence, drawn from both modal shift studies and field experience, converges on a common finding: free fares increase public transport ridership, but attract primarily pedestrians and cyclists, not car drivers. The net impact on GHG emissions is therefore limited in the short term, and may even be counterproductive if the ridership increase implies service degradation without supply reinforcement. This finding is particularly pronounced in large metropolitan areas, where the car is already poorly competitive for many journeys. It should be noted that these conclusions remain methodologically fragile: most available evaluations cannot isolate the effect of free fares from simultaneous reforms, and the absence of a rigorous counterfactual limits the scope of causal conclusions.

The financing question is central and conditions the viability of any free-fare project. The elimination of fare revenues cannot be compensated solely by adjustments to the mobility levy, which has already reached its ceiling in most MOAs. Four alternative mechanisms have been identified: urban congestion charges, land value capture, parking charges, and earmarked national tax revenues. Each presents significant political or practical constraints; it is likely that a combination of these instruments would be required to finance large-scale free fares without undermining network investment capacity. Over the longer term, three orientations for reforming the mobility levy appear promising: transforming it into an instrument of territorial cooperation, adapting its perimeters to the reality of mobility catchment areas, and assigning intermodal resources to regional lead authorities.

Ultimately, free fares are not a universal answer to the challenges of sustainable urban mobility. They are an effective tool for social accessibility in contexts where fare revenues are low and networks are small. They are, however, an insufficient — and potentially counterproductive — tool if introduced at the expense of investment in supply. The genuine lever for decarbonising transport lies more in network expansion and quality,

intermodal complementarity, and car disincentive measures than in pricing alone. Free fares can accompany these transformations — and sometimes catalyse them, as in Dunkirk — but they cannot replace them.

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